

Introduction to Computer and Computer Science

Homework Assignment #10

Due: 12/11 0:00 am

Mortgage & Loan

You might be unfamiliar with the rule of mortgage in modern financial system. But some of you might have some experience in applying the student loan for the tuitions. In this homework, we are going to develop a program, which aims to calculate the monthly payment of mortgage (or loan) of size l at lending rate r for m months. And we are also going to plot some figures for visualizing the outcome of your program.

Enjoy it!

Problem #1: Calculate the monthly payment for fixed rate

Please create a superclass called “**Mortgage**,” which contains the total amount of loan, the lending rate, and how many months for the repayment. When you use this class object in program, it should require these three input parameters as shown below,

Mortgage(l, r, m),

where (l, r, m) represent the total amount of loan, the lending rate (annual), and how many months for the repayment, respectively. We already give you a sample code of a child class “**Fixed**.” Please add some methods in the superclass “**Mortgage**” and use them to calculate the monthly payment. You can test your program by the following cases:

Case 1

$l, r, m = 600000, 0.01, 264$

And the program should return you:

Monthly payment: 2614.4203485197268

Total payment: 690206

Case 2

$l, r, m = 10000000, 0.0175, 360$

And the program should return you:

Monthly payment: 35724.34474172349

Total payment: 12860764

Hint: At least you need 2 additional methods in the superclass “**Mortgage**” so that you can calculate the monthly payment and update the outstanding every month.

Problem #2: Calculate the monthly payment for two different rates

In practical, a mortgage with fixed lending rate is very unusual. The lender always wants to extract the maximum profit from borrower. That makes the lending rate become variable. For example, the lender may set an initial rate for the first 5 years (let's say, 1% for the first 5 years), and then it goes up over time (2% for the rest of years, or gradually increasing by 0.5% every year for the rest of years). In this problem, you are going to deal with a mortgage with one initial rate r_1 and another lending rate r_2 after m_1 months (higher, of course). We already give you a sample code of a child class “**TwoRate**.” Please use them to calculate the monthly payment of mortgage (or loan) of size l at initial rate r_1 for m_1 months, and at lending rate r_2 for the rest $m - m_1$ months. You can test your program by the following cases:

Case 1

$l, r_1, m, r_2, m_1 = 600000, 0.01, 264, 0.025, 60$

And the program should return you:

Monthly payment for the first m_1 months: 2532.8295835813587

Monthly payment for the rest months: 2860.451309390662

Total payment: 735501

Case 2

$l, r_1, m, r_2, m_1 = 10000000, 0.01, 360, 0.025, 60$

And the program should return you:

Monthly payment for the first m_1 months: 32163.952044650043

Monthly payment for the rest months: 38286.924287754904

Total payment: 13415914

Hint: You should override the method that you just create in the superclass “**Mortgage**” in order to calculate the monthly payment under variable-rate in this child class “**TwoRate**.”

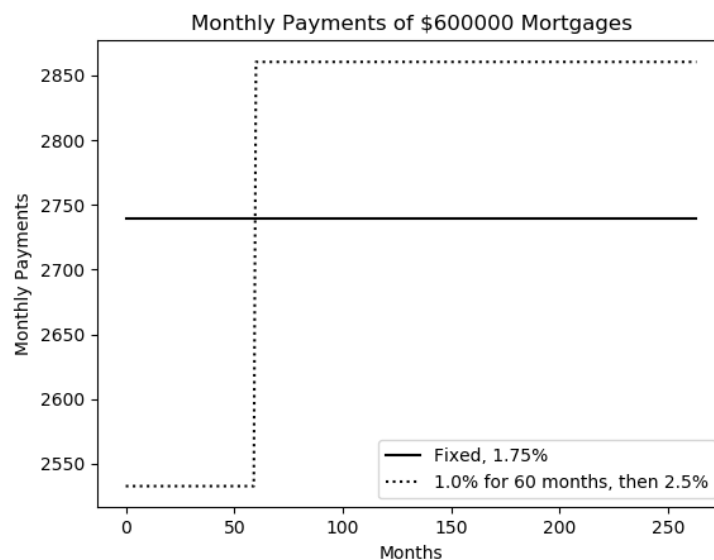
Problem #3: Plot the figures

As we had emphasized the importance of data visualization during the classes, we are going to visualize the result of above questions in this problem. Please try to plot two figures as request:

Figure 1 Monthly Payment of \$\$\$ Mortgages

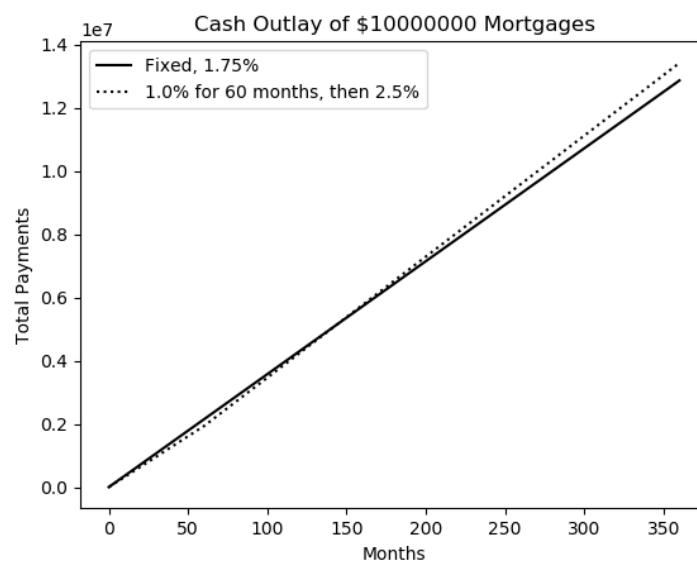
For example, let's assume $l, r_1, m, r_2, m_1 = 600000, 0.01, 264, 0.025, 60$

And we should acquire a figure as shown below:

**Figure 2 Cash Outlay of \$\$\$ Mortgages**

For example, let's assume $l, r_1, m, r_2, m_1 = 10000000, 0.01, 360, 0.025, 60$

And we should acquire a figure as shown below:



Hint: All of the numbers in the figure should be automatically changed under different situations. That means, you should try to add additional methods for plotting the corresponding figures in your class object.

Monthly payment:

Month	Outstanding
0	l
1	$l(1 + r) - pay$
2	$[l(1 + r) - pay](1 + r) - pay$
3	$l(1 + r)^3 - pay(1 + r)^2 - pay(1 + r) - pay$
...	...
m	$??? = 0$

Good luck and enjoy it!

Please be aware, the teaching assistants will use “import” command to import your program and execute some testing code to examine your program. **The name of your functions, class object, including the name of corresponding methods, should be exactly as same as the name we shown here.** Otherwise it will pop out an error and terminate the testing procedure. In that case, your homework will be considered to be failed. **Please double check your program carefully.**

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```

6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |

```

Please save your code as a “.py” file, where the file name should follow this format:

108A_hw#?_YourID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.